

Alligator vs Crocodile Image Classification – Conclusion

1. Models Tried

- Custom CNN (3 Convolutional Layers) – Validation Accuracy: 60.81%
- MobileNetV2 (Transfer Learning) – Validation Accuracy: 63.92%
- EfficientNetB0 (Transfer Learning) – Validation Accuracy: 53.51%

2. Best Performing Model

MobileNetV2 achieved the highest validation accuracy of 63.92%. It performed better than the Custom CNN and EfficientNetB0 due to transfer learning from ImageNet, which helped extract better feature representations for visually similar classes like alligators and crocodiles.

3. Final Performance Metrics (Selected Model – MobileNetV2)

Metric	Score
Best Validation Accuracy	63.92%
Custom CNN Validation Accuracy	60.81%
EfficientNetB0 Validation Accuracy	53.51%

4. Evaluation Strategy

- Accuracy used as primary evaluation metric.
- Validation accuracy monitored to detect overfitting.
- Comparison across multiple architectures performed to select best model.

5. Conclusion

Among the tested architectures, MobileNetV2 delivered the best performance with 63.92% validation accuracy. The results indicate that transfer learning provides improved generalization compared to a custom CNN for small and visually similar datasets.